

assignment6-key

October 8, 2025

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[1]: # Install if needed:
# !pip install geopandas matplotlib folium pandas

import geopandas as gpd
import matplotlib.pyplot as plt
import pandas as pd
import folium

#
# 1. Load Natural Earth Low-Resolution Data
#
world = gpd.read_file(gpd.datasets.get_path("naturalearth_lowres"))

# Filter to Australia
aus = world[world["name"] == "Australia"]
print(aus)

#
# 2. Static map of Australia
#
fig, ax = plt.subplots(figsize=(8, 6))
aus.plot(ax=ax, color="lightgreen", edgecolor="black")
ax.set_title("Map of Australia", fontsize=14)
ax.set_axis_off()
plt.show()

#
# 3. Add Australian capital cities (no iterrows)
#
capitals = pd.DataFrame({
    "city": ["Sydney", "Melbourne", "Brisbane", "Adelaide",
            "Perth", "Hobart", "Canberra", "Darwin"],
    "lat": [-33.8688, -37.8136, -27.4698, -34.9285,
            -31.9505, -42.8821, -35.2809, -12.4634],
    "lon": [151.2093, 144.9631, 153.0251, 138.6007,
            115.8605, 147.3272, 149.1300, 130.8456]
})
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fig, ax = plt.subplots(figsize=(8, 6))
aus.plot(ax=ax, color="lightblue", edgecolor="black")
plt.scatter(capitals["lon"], capitals["lat"], color="red", s=25)

# Add city labels using basic indexing
for i in range(len(capitals)):
    x = capitals["lon"][i]
    y = capitals["lat"][i]
    name = capitals["city"][i]
    plt.text(x + 0.5, y + 0.5, name, fontsize=8)

ax.set_title("Australia with Capital Cities")
ax.set_axis_off()
plt.show()

#
# 4. Interactive map using folium (no iterrows)
#
m = folium.Map(location=[-25, 135], zoom_start=4)

folium.GeoJson(
    aus,
    name="Australia",
    style_function=lambda x: {
        "fillColor": "#66c2a5",
        "color": "black",
        "weight": 0.8,
        "fillOpacity": 0.6
    },
    tooltip=folium.features.GeoJsonTooltip(fields=["name"], aliases=["Country:
↵"])
).add_to(m)

m

#
# 5. Interactive map using folium (no iterrows)
#
m = folium.Map(location=[-25, 135], zoom_start=4)

folium.GeoJson(
    aus,
    name="Australia",
    style_function=lambda x: {
        "fillColor": "#66c2a5",
        "color": "black",

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        "weight": 0.8,
        "fillOpacity": 0.6
    },
    tooltip=folium.features.GeoJsonTooltip(fields=["name"], aliases=["Country:
↵"]),
).add_to(m)

# Add capital cities to map (loop without iterrows)
for i in range(len(capitals)):
    lat = capitals["lat"][i]
    lon = capitals["lon"][i]
    name = capitals["city"][i]

    folium.CircleMarker(
        location=[lat, lon],
        radius=4,
        color="red",
        fill=True,
        fill_opacity=0.9,
        tooltip=name
    ).add_to(m)

m.save("australia_map.html")

m

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    pop_est continent      name iso_a3  gdp_md_est  \
137  23232413  Oceania  Australia    AUS    1189000.0

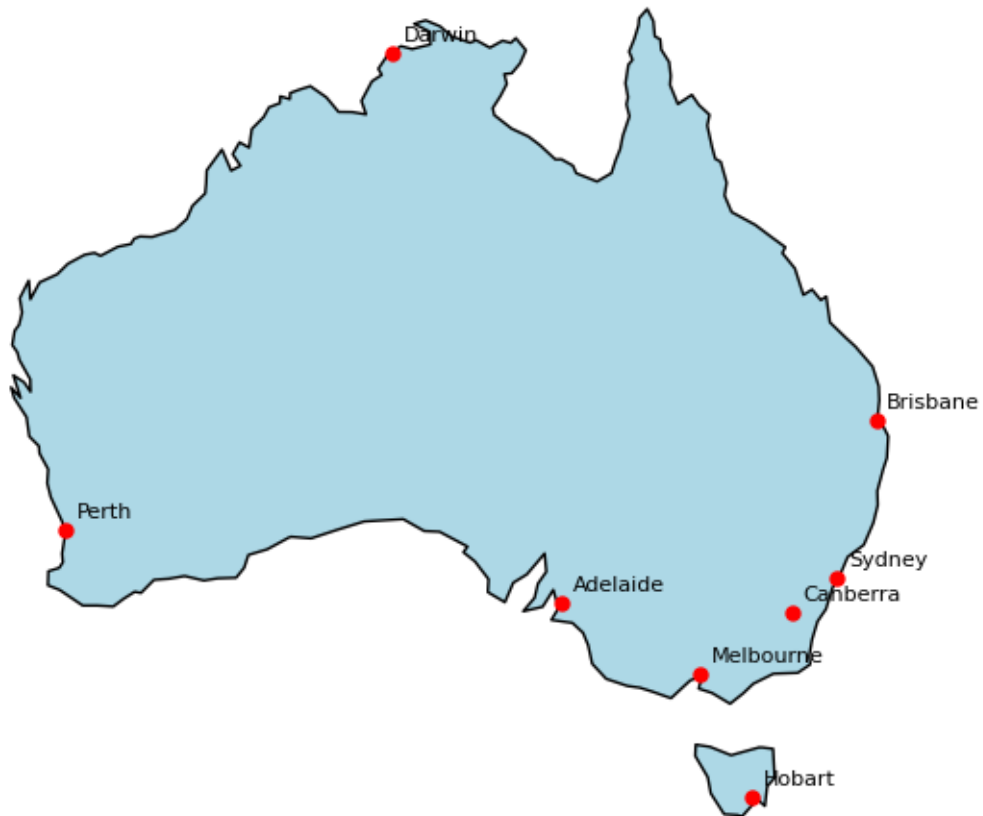
                                geometry
137  MULTIPOLYGON (((147.68926 -40.80826, 148.28907...

```

Map of Australia



Australia with Capital Cities



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[1]: <folium.folium.Map at 0x15b9932b0>
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